

X-91 Flight-Control Approval Data Pack

Candidate v0.4 - Qualified Review Required

Civilian non-weaponized demonstrator

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Approval statement

Disposition: READY FOR QUALIFIED REVIEW, NOT APPROVED FOR FLIGHT. This package defines the evidence, configuration and pass/fail gates needed for a qualified flight-controls engineer and test pilot to issue a restricted initial flight release. It does not grant regulatory authorization, airworthiness, fabrication release, or permission to operate.

Scope is a civilian, non-weaponized demonstrator. Payload release, targeting, person tracking, pursuit, combat logic and autonomous mission execution are excluded. Initial approval is limited to MANUAL, FBWA and RTL under direct visual-line-of-sight control.

Controlled configuration

Element	Approval baseline	Configuration control
Flight controller	Holybro Pixhawk 6X	Record serial number, ha
Flight software	ArduPlane stable release	Reviewer freezes exact s
Navigation	External GNSS/compass, barometer and calibrated digital airspeed	Record models, firmware
Pilot command	Independent RC link with physical mode switch and immediate MANUAL selection	Demonstrate command l
Power	Two independent regulated avionics inputs; propulsion BECs electrically excluded	Brownout, single-feed lo
Actuation	Two KST X10 V8.0 elevon servos at regulated 7.4 V	Independent output char
Propulsion	Two independently monitored EDF branches	Common throttle comm
Logging	Onboard high-rate log plus independent test recorder	Parameters, firmware id

Approval envelope

- Maximum gross mass: 4.50 kg; CG: 24% MAC nominal, allowable initial range 23-25% MAC after physical weighing.
- Initial airspeed envelope: 1.3 times measured stall speed through 25 m/s. Expansion above 25 m/s requires correlated GVT/flutter clearance; never exceed the lesser of the analyst-approved limit or 45 m/s.
- Bank command: +/-30 deg; pitch command: +15/-10 deg; pitch rate <=45 deg/s; roll rate <=90 deg/s.
- Wind: <=5 m/s steady, <=2 m/s gust spread; dry daylight; prepared field; no flight over people or occupied structures.
- Initial altitude: local rule-compliant, with minimum recovery height selected by test pilot and maximum geofence altitude/radius set for the site.
- Maximum flight duration: value derived from instrumented propulsion/endurance test with >=30% battery reserve on landing.

Requirements and verification

ID	Requirement	Verification
FC-001	Exact hardware, firmware and parameters are uniquely identified.	Inspection
FC-002	Manual control remains available without GNSS, airspeed, telemetry or GCS.	HITL/ground
FC-003	Any single avionics power-feed loss causes no reboot or actuator transient.	Bench
FC-004	Sensor installation is free of unacceptable vibration, magnetic and pressure errors.	Ground/taxi

ID	Requirement	Verification
FC-005	Command limits prevent structural/envelope exceedance in FBWA.	SITL/HITL
FC-006	RC loss produces the approved RTL sequence and reacquisition restores pilot authority predictably.	SITL/HITL/rang
FC-007	Geofence breach produces the approved return behavior without leaving the containment volume.	SITL/HITL
FC-008	One EDF degradation/failure does not produce unrecoverable yaw/roll.	6-DOF/HITL/tet
FC-009	Servo authority, speed and thermal margin cover the approved envelope.	Bench/load
FC-010	State estimation degrades safely under sensor faults.	SITL/HITL
FC-011	All safety-critical events and commands are reconstructable.	Log review
FC-012	Initial flights cannot enter autonomous mission execution.	Inspection/test

Model acceptance before control-law approval

The current 6-DOF derivatives are assumptions and cannot support gain approval. Replace them with:

1. Measured mass, CG and full inertia tensor for minimum and maximum flight mass.
2. Aerodynamic coefficient tables versus angle of attack, sideslip, control deflection and propulsion state, including post-stall uncertainty bounds.
3. Propulsion thrust and lag versus command, airspeed and battery voltage, including left/right mismatch.
4. Servo position, rate limit, deadband, backlash, bus-voltage dependence and load response.
5. Correlated structural modes from GVT, with flight-control sampling/filter frequencies checked against elastic modes.

The nonlinear model passes only if trim residuals are within 2% weight and 0.5 deg control, predicted static margin agrees with glide/ground evidence within 2% MAC, and measured doublet responses fall inside the model's 95% uncertainty bounds. Control gains cannot be finalized before correlation.

SITL and HITL campaign

Use the frozen ArduPlane binary and controlled parameter file. Run nominal cases and at least 200 Monte Carlo trials per safety-critical scenario across mass, CG, aerodynamic uncertainty, wind, sensor bias, battery voltage, servo rate and propulsion mismatch.

Scenarios: takeoff/launch transient; pilot doublets; stall approach and recovery; minimum/maximum CG; 5 m/s cross-wind; gusts; RC loss/recovery; GCS loss; GNSS dropout/jump; compass bias; airspeed freeze/blockage/bias; barometer drift; one power-feed loss; one servo slow/jammed at neutral/offset; one EDF 10/25/50/100% loss; fence breach at every boundary; RTL from minimum energy; CPU load and logging saturation.

Pass criteria: no departure, sustained oscillation, command reversal, geofence escape, reboot or structural-envelope exceedance; pilot override always works; required annunciation occurs; terminal states are controlled landing/approved loiter or manual recovery. Every failure is dispositioned before approval.

Ground and restrained tests

1. Configuration audit and independent control-direction check.
2. Avionics power interruption, brownout and servo-load test while logging.
3. Vibration survey with each EDF alone and together through the full RPM range; inspect notch/filter separation from structural modes.
4. Pitot leak, zero, scale and blockage tests; compass interference test at every throttle setting.
5. RC/GCS range and antenna-orientation tests; verify failsafe timing and manual reacquisition.
6. Restrained asymmetric-thrust detection using controlled RPM/current reduction on one branch.
7. Hardware-in-loop test with the exact installed controller, receiver, sensors and servo loads.

Restricted flight-test cards

Card	Objective	Entry gate	Maximum task
FT-01	Unpowered glide trim	Structural proof, CG and manual control signed	Straight glide, small pitch corrections, no
FT-02	Powered straight flight	Propulsion, vibration and FT-01 pass	MANUAL only, ≤ 18 m/s, shallow turn
FT-03	Stabilized-mode evaluation	Correlated manual response and HITL pass	FBWA, 5 deg pitch and 15 deg bank step
FT-04	Envelope identification	FT-03 stable and model updated	Small doublets, stall approach at safe he

Card	Objective	Entry gate	Maximum task
FT-05	RTL demonstration	RC-loss/geofence HITL 20/20 pass	Pilot-commanded RTL with immediate M

Each card requires test pilot, flight-test engineer and safety observer; weather/site go-no-go; abort plan; telemetry and onboard logging; post-flight inspection; data review and written authorization before the next card.

Approval signatures

Role	Required finding	Name / credential
Flight-controls engineer	Requirements, model correlation, gain/limit and V&V evidence acceptable	_____
Aeroelastic/structures reviewer	Approved control envelope is inside structural and flutter clearance	_____
Electrical/propulsion reviewer	Power, actuator and asymmetric-thrust protections acceptable	_____
Flight-test pilot	Test cards, abort criteria, handling risks and site limits accepted	_____
Configuration manager	Hardware/software/parameter baseline and evidence hashes complete	_____

Approval, if granted, is limited to the identified airframe serial number, mass/CG range, firmware hash, parameter hash, hardware configuration, site and envelope. Any change affecting geometry, mass distribution, servo linkage, propulsion, sensors, flight software or parameters suspends approval pending impact review.

References

- ArduPilot Pixhawk 6X: <https://ardupilot.org/plane/docs/common-holybro-pixhawk6X.html>
- ArduPilot SITL: <https://ardupilot.org/dev/docs/using-sitl-for-ardupilot-testing.html>
- ArduPilot airspeed sensor guidance: <https://ardupilot.org/plane/docs/airspeed.html>
- ArduPilot advanced failsafe configuration: <https://ardupilot.org/plane/docs/advanced-failsafe-configuration.html>
- ArduPilot parameter reference: <https://ardupilot.org/plane/docs/parameters.html>
- FAA educational/recreational operations: https://www.faa.gov/uas/educational_users
- FAA Part 107 waivers: https://www.faa.gov/uas/commercial_operators/part_107_waivers